

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA0706 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

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Code of Conduct:

I _____V.Lasya(192524186)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V.lasya

Signature of the student:

1 . A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = 150 MB = 150,000,000 bytes
- Bandwidth = 50 Mbps
- Transport Layer segment size = 1500 bytes
- Data Link Layer overhead = 40 bytes/frame

1. Number of Segments

$$\begin{aligned}\text{Number of Segments} &= \frac{\text{File Size}}{\text{Segment Size}} \\ &= \frac{150,000,000}{1500} = 100,000\end{aligned}$$

Answer: 100,000 segments

2. Number of Frames

Each segment is encapsulated into one frame.

$\text{Frames} = 100,000$

Answer: 100,000 frames

3. Total Data Link Layer Overhead

Each frame adds 40 bytes.

$$\begin{aligned}100,000 \times 40 &= 4,000,000 \text{ bytes} \\ 4,000,000 \text{ bytes} &= 4 \text{ MB}\end{aligned}$$

Answer: 4,000,000 bytes (4 MB) of overhead

4. Total Transmission Time

Total data transmitted

$$150,000,000 + 4,000,000 = 154,000,000 \text{ bytes}$$

Convert to bits:

$$154,000,000 \times 8 = 1,232,000,000 \text{ bits}$$

Bandwidth:

$$50 \text{ Mbps} = 50,000,000 \text{ bits/sec}$$

Transmission time:

$$\frac{1,232,000,000}{50,000,000} = 24.64 \text{ seconds}$$

Answer: 24.64 seconds

Final Answers

- Number of segments: 100,000
- Number of frames: 100,000
- Data Link Layer overhead: 4,000,000 bytes (4 MB)
- Transmission time: 24.64 seconds

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
 - Frame size = 1024 bytes
 - Total frames = 48
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1. Number of Transmission Windows

$$\begin{aligned} \text{Number of Windows} &= \frac{\text{Total Frames}}{\text{Window Size}} \\ &= \frac{48}{6} = 8 \end{aligned}$$

Answer: 8 transmission windows

2. Total Number of Retransmissions

Given that one frame is lost in every window:

- Number of windows = 8
- Lost frames = 1 per window

$$8 \times 1 = 8$$

Answer: 8 retransmissions

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes

- IP header size = 20 bytes per fragment

1. Maximum Data per Fragment

Since each fragment requires a 20-byte IP header:

$$\text{Data per fragment} = 1500 - 20 = 1480 \text{ bytes}$$

2. Number of Fragments

$$\begin{aligned}\text{Number of Fragments} &= \left\lceil \frac{12000}{1480} \right\rceil \\ &= \lceil 8.108 \rceil = 9\end{aligned}$$

Answer: 9 fragments

3. Total Header Overhead

Each fragment has a 20-byte IP header.

$$9 \times 20 = 180 \text{ bytes}$$

Answer: 180 bytes

Final Answers

- Original IP packet size: 12,000 bytes
- MTU: 1500 bytes
- Data per fragment: 1480 bytes
- Number of fragments: 9
- Total IP header overhead: 180 bytes

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
- Each frame carries = 1024 bytes
- Total frames transmitted = 48

1. Number of Transmission Windows

$$\begin{aligned}\text{Number of Windows} &= \frac{\text{Total Frames}}{\text{Window Size}} \\ &= \frac{48}{6} = 8\end{aligned}$$

Answer: 8 transmission windows

2. Total Number of Retransmissions

Since one frame is lost in every window:

$$8 \text{ windows} \times 1 \text{ lost frame/window} = 8$$

Answer: 8 retransmissions

Final Answers

- Sliding window size: 6 frames
- Frame size: 1024 bytes
- Total frames: 48
- Transmission windows required: 8
- Total retransmissions: 8 frames

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given:

- Video file size = 600 MB
 - Checkpoint interval = 60 MB
 - Failure occurs after = 390 MB
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1. Number of Checkpoints Created Before Failure

Checkpoints are inserted every 60 MB.

$$\frac{390}{60} = 6.5$$

Only completed checkpoints are counted.

Checkpoints are created at:

- 60 MB
- 120 MB
- 180 MB
- 240 MB
- 300 MB
- 360 MB

Answer: 6 checkpoints

2. Data to be Retransmitted

The last successful checkpoint is at 360 MB.

Failure occurs at 390 MB.

$$390 - 360 = 30 \text{ MB}$$

After recovery, transmission resumes from the 360 MB checkpoint.

Answer: 30 MB must be retransmitted

Final Answers

- Total checkpoints before failure: 6
- Last checkpoint: 360 MB
- Data to be retransmitted: 30 MB

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original file size = 900 MB
- Compression = 40%
- Encryption overhead = 5% of compressed size

1. Compressed File Size

A 40% compression means the file size becomes 60% of the original.

$$\text{Compressed Size} = 900 \times 0.60 = 540 \text{ MB}$$

Answer: 540 MB

2. Encrypted File Size

Encryption adds 5% overhead to the compressed size.

$$\text{Overhead} = 540 \times 0.05 = 27 \text{ MB}$$

$$\text{Encrypted Size} = 540 + 27 = 567 \text{ MB}$$

Answer: 567 MB

3. Total Percentage Reduction

Reduction in size:

$$900 - 567 = 333 \text{ MB}$$

Percentage reduction:

$$\frac{333}{900} \times 100 = 37\%$$

Answer: 37% reduction

Final Answers

- Original file size: 900 MB

- Compressed file size: 540 MB
- Encrypted file size: 567 MB
- Total percentage reduction: 37%

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given:

- Total data = 3 GB = 3000 MB
- Throughput = 120 Mbps
- Request size = 3 MB

1. Total Number of Requests

$$\begin{aligned}\text{Number of Requests} &= \frac{\text{Total Data}}{\text{Request Size}} \\ &= \frac{3000}{3} = 1000\end{aligned}$$

Answer: 1000 requests

2. Total Transmission Time

Convert data to megabits:

$$\begin{aligned}3 \text{ GB} &= 3000 \text{ MB} \\ 3000 \times 8 &= 24,000 \text{ Mb}\end{aligned}$$

Transmission time:

$$\text{Time} = \frac{24,000}{120} = 200 \text{ seconds}$$

Answer: 200 seconds

Final Answers

- Total data: 3 GB
- Request size: 3 MB
- Number of requests generated: 1000
- Transmission time: 200 seconds

8. A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the

communication process.

Given:

- Application Layer delay = 4 ms
- Transport Layer delay = 3 ms
- Network Layer delay = 5 ms
- Data Link Layer delay = 2 ms
- Physical Layer delay = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

1. Total Processing Delay

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Answer: 15 ms

2. Total End-to-End Delay

$$\begin{aligned} \text{Total Delay} &= \text{Processing Delay} + \text{Propagation Delay} + \text{Transmission Delay} \\ &= 15 + 18 + 12 = 45 \text{ ms} \end{aligned}$$

Answer: 45 ms

Final Answers

- Processing delay: 15 ms
- Propagation delay: 18 ms
- Transmission delay: 12 ms
- Total end-to-end delay: 45 ms

9 .A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Protocol overhead per frame = 80 bytes

1. Useful Data per Frame

Since each frame has 80 bytes of overhead:

$$\text{Useful data per frame} = 1600 - 80 = 1520 \text{ bytes}$$

2. Total Number of Frames Required

$$\begin{aligned}\text{Number of Frames} &= \left\lceil \frac{80,000,000}{1520} \right\rceil \\ &= \lceil 52631.58 \rceil = 52,632\end{aligned}$$

Answer: 52,632 frames

3. Total Protocol Overhead

$$\begin{aligned}52,632 \times 80 &= 4,210,560 \text{ bytes} \\ &= 4.21 \text{ MB (approximately)}\end{aligned}$$

Answer: 4,210,560 bytes ($\approx$ 4.21 MB)

Final Answers

- Useful data per frame: 1520 bytes
- Number of frames required: 52,632
- Total protocol overhead: 4,210,560 bytes ($\approx$ 4.21 MB)

10.A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given:

- Original file size = 240 MB = 240,000,000 bytes
 - Compression = 30%
 - Segment size = 1400 bytes
 - Data Link Layer overhead = 60 bytes per frame
-

1. Compressed File Size

A 30% compression means the file becomes 70% of its original size.

$$\begin{aligned}\text{Compressed Size} &= 240 \times 0.70 = 168 \text{ MB} \\ 168 \text{ MB} &= 168,000,000 \text{ bytes}\end{aligned}$$

Answer: 168 MB

2. Total Number of Segments

$$\text{Number of Segments} = \frac{168,000,000}{1400} = 120,000$$

Answer: 120,000 segments

3. Total Protocol Overhead

Each frame adds 60 bytes of overhead.

$$120,000 \times 60 = 7,200,000 \text{ bytes}$$
$$7,200,000 \text{ bytes} = 7.2 \text{ MB}$$

Answer: 7,200,000 bytes (7.2 MB)

4. Final Amount of Data Transmitted Across the Physical Layer

$$168,000,000 + 7,200,000 = 175,200,000 \text{ bytes}$$
$$175,200,000 \text{ bytes} = 175.2 \text{ MB}$$

Answer: 175.2 MB

Final Answers

- Compressed file size: 168 MB
- Total segments transmitted: 120,000
- Total protocol overhead: 7,200,000 bytes (7.2 MB)
- Final data transmitted across the Physical Layer: 175.2 MB